Instrumental Variable Estimation in Telco Retention

Overcoming Unmeasured Confounding via Exogenous Triangulation

1. The Unmeasured Confounding Challenge in Churn Modeling

In telecommunications retention operations, evaluating whether a retargeting bill discount (treatment **A** = 1 vs 0) reduces 90-day subscriber churn (outcome **Y**) relies heavily on observational customer logs. Traditional methods like Inverse Probability Weighting (IPW) or Outcome Regression require the **Conditional Exchangeability** assumption ($Y^a \perp A \mid L$).

However, in real-world retention workflows, key drivers of churn are frequently **unmeasured in enterprise data warehouses (U)** — such as unrecorded customer calls to competitors, latent dissatisfaction, or private intention to switch providers. Because dissatisfied subscribers actively seek retention discounts, treatment **A** is confounded by **U**. When U exists, both IPW and regression estimates become fundamentally biased.

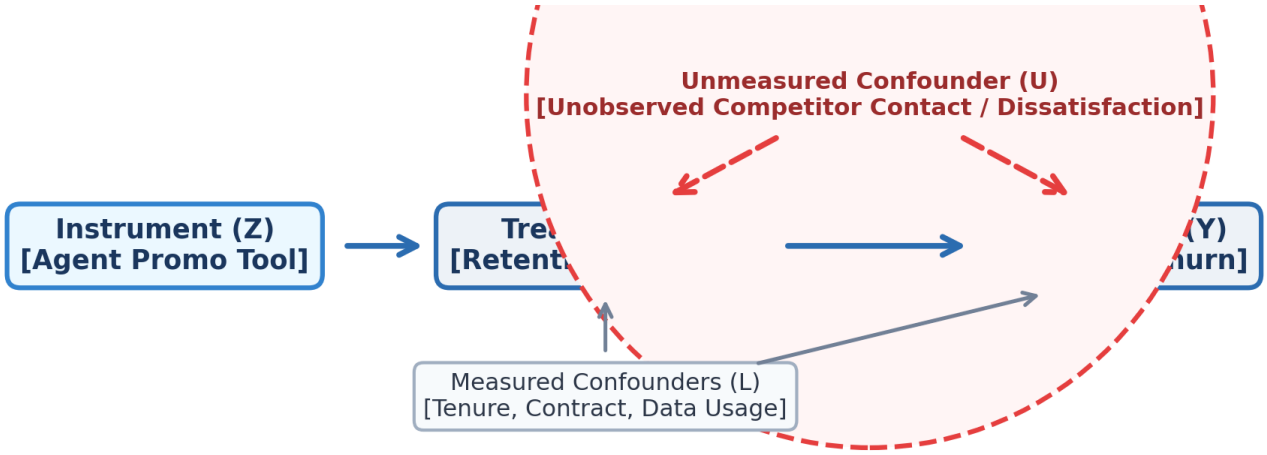


Figure 16.1: Structural Causal Model with Unmeasured Confounder (U) and Instrumental Variable (Z).

2. The Three Core Instrumental Variable (IV) Conditions

An **Instrumental Variable (Z)** is an exogenous factor that affects treatment assignment **A** without directly influencing the outcome **Y** or sharing unmeasured confounders with **Y**.

Condition & Mathematical Definition	Operational Meaning in Telco Retention	Potential Failure Mode / Threat
1. Relevance (Instrument-Treatment Link) $Z \not\perp A \mid L$	The instrument Z must strongly predict whether a subscriber receives the retention discount A (e.g., call center agent routed with promo tool access).	**Weak Instrument Bias:** If Z has a weak effect on A (F-statistic < 10), IV variance explodes and finite-sample bias increases.
2. Exclusion Restriction $Y^{a,z} = Y^a$	The instrument Z affects 90-day churn Y **ONLY** through its effect on treatment A. There is no direct causal arrow Z → Y.	**Direct Path Violation:** Promo agents also offer general service advice or friendly support that directly reduces churn independently of the discount.

Condition & Mathematical Definition	Operational Meaning in Telco Retention	Potential Failure Mode / Threat
3. Instrument Unconfoundedness $Z \perp Y^a \mid L$	The instrument Z shares no measured or unmeasured confounders with outcome Y (i.e. Z is effectively randomized conditional on L).	**Confounded Instrument:** Call routing algorithms selectively route high-value or angry callers to promo-equipped agents.

3. Compliance Principal Strata & Local Average Treatment Effect (LATE)

In the presence of unmeasured confounding, an instrumental variable generally cannot identify the Average Treatment Effect (ATE) for the entire population without extreme homogeneity assumptions. Instead, under the **Monotonicity Assumption** (no defiers: $A^{z=1} \geq A^{z=0}$), IV identifies the **Complier Average Causal Effect (CACE) / Local Average Treatment Effect (LATE)**.

Principal Compliance Classes in Telco Retention:
• **Compliers ($A^{z=1}=1, A^{z=0}=0$):** Subscribers who accept/receive the discount if routed to a promo agent (Z=1), but NOT if routed to a standard agent (Z=0).
• **Always-Takers ($A^{z=1}=1, A^{z=0}=1$):** Aggressive subscribers who demand and obtain discounts regardless of agent assignment.
• **Never-Takers ($A^{z=1}=0, A^{z=0}=0$):** Subscribers who reject retention offers regardless of agent assignment.
LATE Interpretation: IV measures the causal effect of retargeting discounts specifically among Compliers.

4. Mathematical Estimators & Two-Stage Least Squares (2SLS)

When Z, A, and Y are measured, the classic **Wald Estimator** computes the ratio of the Intention-to-Treat (ITT) effect on the outcome to the ITT effect on treatment engagement:

$$\textbf{Wald Estimator:} \quad \theta_{IV} = [\, E[Y \mid Z = 1] - E[Y \mid Z = 0] \,] \,/\, [\, E[A \mid Z = 1] - E[A \mid Z = 0] \,]$$

To incorporate baseline measured covariates L (e.g. tenure, contract type) and improve efficiency, we deploy the **Two-Stage Least Squares (2SLS)** algorithm:

Two-Stage Least Squares (2SLS) Workflow:
Stage 1 (Treatment Prediction): Fit linear model predicting discount receipt A using instrument Z and covariates L:

$$\hat{A} = \gamma_0 + \gamma_1 Z + \gamma_2 L$$

Stage 2 (Outcome Model): Regress 90-day churn Y on predicted treatment values Â and covariates L:

$$Y = \beta_0 + \beta_{IV} \hat{A} + \beta_2 L$$

The coefficient β_{IV} yields the consistent LATE estimate of retention discount efficacy.

5. Empirical Telco Results & Methodological Comparison

Method / Estimator	Assumed Confounding Status	Estimated Churn Effect (ATE/LATE)	Methodological Risk / Evaluation
Naive Comparison	Ignores all confounding (U, L).	**-8.8%** (Exaggerated Benefit)	Confounded by selective discount targeting of high-churn callers.
IP Weighting (Chapter 12)	Assumes No Unmeasured Confounding ($U = ∅$).	**-2.1%** (Biased by U)	Removes bias from L, but fails due to unmeasured customer churn intent U.
2SLS IV Estimator (Chapter 16)	Robust to Unmeasured Confounding (U).	**-4.5% (LATE among Compliers)**	Consistent causal estimate for Complier segment; requires valid Z.

6. Practical Production Guidance for Telco Data Teams

• **Always verify instrument strength:** Run first-stage F-tests. If F < 10, do not use standard IV; deploy weak-instrument robust confidence intervals (Anderson-Rubin).

• **Audit Exclusion Restriction:** Ensure call center agents with promo tool access do not systematically provide other unmeasured incentives (e.g. priority queueing or technical support).

• **Clarify target population:** Recognize that IV identifies LATE (Compliers), not ATE. If business leadership requires population-wide ATE, pair IV with sensitivity analyses for unmeasured confounding.